



## Atividade pratica 04: Atividade de laboratório – Consulta DNS

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### **Parte 1: Introdução.**

#### **Ferramenta 1: nslookup**

1. Resolver um nome e identificar servidor que respondeu:

```
rizzonly@DESKTOP-G55DH5D:~$ nslookup www.google.com
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.250.78.164
Name:   www.google.com
Address: 2800:3f0:4004:80d::2004
```

Nome resolvido: google.com, com IP associado 142.250.78.164 //

2800:3f0:4004:80d::2004

Servidor que respondeu: 10.255.255.254

2. Consultar tipo MX (servidores de e-mail):

```
rizzonly@DESKTOP-G55DH5D:~$ nslookup -type=MX gmail.com
Server:      10.255.255.254
Address:     10.255.255.254#53

Non-authoritative answer:
gmail.com    mail exchanger = 30 alt3.gmail-smtp-in.l.google.com.
gmail.com    mail exchanger = 10 alt1.gmail-smtp-in.l.google.com.
gmail.com    mail exchanger = 20 alt2.gmail-smtp-in.l.google.com.
gmail.com    mail exchanger = 40 alt4.gmail-smtp-in.l.google.com.
gmail.com    mail exchanger = 5 gmail-smtp-in.l.google.com.

Authoritative answers can be found from:
```

3. Trocar o servidor DNS (ex.: Cloudflare):

```
rizzonly@DESKTOP-G55DH5D:~$ nslookup www.unb.br 1.1.1.1
Server:      1.1.1.1
Address:     1.1.1.1#53

Non-authoritative answer:
Name:   www.unb.br
Address: 164.41.102.70
```

## Parte 2 – Explorando mais campos

### Ferramenta 2: dig

1. Consultar registros A e observar TTL:

```
rizzonly@DESKTOP-G55DH5D:~$ dig www.unb.br

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> www.unb.br
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 30424
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
;; QUESTION SECTION:
;www.unb.br.                IN      A

;; ANSWER SECTION:
www.unb.br.                5       IN      A      164.41.102.70

;; Query time: 12 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Fri Oct 03 16:11:18 -03 2025
;; MSG SIZE rcvd: 55
```

Registro A: 164.41.102.70

TTL: 5

2. Consultar IPv6 (AAAA) e saída curta:

```
rizzonly@DESKTOP-G55DH5D:~$ dig +short www.google.com AAAA
2800:3f0:4004:813::2004
```

3. Consultar registros NS:

```
rizzonly@DESKTOP-G55DH5D:~$ dig unb.br NS

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> unb.br NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 15006
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
;; QUESTION SECTION:
;unb.br.                    IN      NS

;; ANSWER SECTION:
unb.br.                    3510    IN      NS      ns2.gigacandanga.net.br.
unb.br.                    3510    IN      NS      dns2.unb.br.
unb.br.                    3510    IN      NS      dns3.unb.br.
unb.br.                    3510    IN      NS      dns1.unb.br.
unb.br.                    3510    IN      NS      ns.gigacandanga.net.br.

;; Query time: 15 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Fri Oct 03 16:16:54 -03 2025
;; MSG SIZE rcvd: 144
```

4. Consultar CNAME:

```
rizzonly@DESKTOP-G55DH5D: $ dig mail.google.com CNAME

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> mail.google.com CNAME
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 3149
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
;; QUESTION SECTION:
;mail.google.com.                IN      CNAME

;; AUTHORITY SECTION:
google.com.                      60      IN      SOA     ns1.google.com. dns-admin.google.com. 813711196 900 900 1800
60

;; Query time: 95 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Fri Oct 03 16:18:10 -03 2025
;; MSG SIZE rcvd: 94
```

## 5. Consultar usando servidor publico (Google DNS):

```
rizzonly@DESKTOP-G55DH5D: $ dig www.globo.com @8.8.8.8

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> www.globo.com @8.8.8.8
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 26752
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.globo.com.                  IN      A

;; AUTHORITY SECTION:
globo.com.                      1800    IN      SOA     ns01.globo.com. fapesp.corp.globo.com. 2025100302 10800 3600
604800 86400

;; Query time: 48 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Fri Oct 03 16:22:12 -03 2025
;; MSG SIZE rcvd: 96
```

## Parte 3 – Hierarquia e interação

### Ferramenta 3: dig +trace:

#### 1. Seguir a cadeia de resolução:

```
rizzonly@DESKTOP-G55DH5D:~$ dig +trace www.unb.br
;; communications error to 10.255.255.254#53: timed out
;; communications error to 10.255.255.254#53: timed out
;; communications error to 10.255.255.254#53: timed out

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> +trace www.unb.br
;; global options: +cmd
;; no servers could be reached
```

Nessa tentativa houve falha, então eu utilizei um servidor DNS publico conhecido (google). Usando o comando: dig +trace @8.8.8.8 [www.unb.br](http://www.unb.br).

```

rizzonly@DESKTOP-G55DH5D: $ dig +trace @8.8.8.8 www.unb.br

<<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> +trace @8.8.8.8 www.unb.br
(1 server found)
;; global options: +cmd
.      87203  IN      NS      m.root-servers.net.
.      87203  IN      NS      h.root-servers.net.
.      87203  IN      NS      j.root-servers.net.
.      87203  IN      NS      g.root-servers.net.
.      87203  IN      NS      e.root-servers.net.
.      87203  IN      NS      f.root-servers.net.
.      87203  IN      NS      c.root-servers.net.
.      87203  IN      NS      k.root-servers.net.
.      87203  IN      NS      d.root-servers.net.
.      87203  IN      NS      l.root-servers.net.
.      87203  IN      NS      i.root-servers.net.
.      87203  IN      NS      a.root-servers.net.
.      87203  IN      NS      b.root-servers.net.
.      87203  IN      RRSIG   NS 8 0 518400 20251016170000 20251003160000 61809 . 4sBNZ+dgOtt3W5V
ilu3sSxMIjH7zbWCONNH0sAznS/h099dWuM8zVbL 4mGzkl03qITnm5TGRCNXEFQEqJVeUfYFPEDfVZsqV7NJprB9AsopokCf 2y8zfUyLPDImYLGU
RCydk4eL10dkQYbS3sJ+H2a540sZodP2HWRwDVGs H1kuARi1qE0mX+uND4n0IRxuw2UsarKhaeWEC1U0A/4TKuItUQcaekX N8NqA6/O3qUE0cT4M
4vjkuY16MQds6MDnYS3rHQzHXp55zV/NvJHnv7I /b0A5EihgU/XKyFFZbpJ62/rhxKdvMURPW5H/UVE7QcLVAp1qm6bo3ZC DqmHIw==
;; Received 525 bytes from 8.8.8.8#53(8.8.8.8) in 43 ms

br.      172800  IN      NS      a.dns.br.
br.      172800  IN      NS      b.dns.br.
br.      172800  IN      NS      c.dns.br.
br.      172800  IN      NS      d.dns.br.
br.      172800  IN      NS      e.dns.br.
br.      172800  IN      NS      f.dns.br.
br.      86400  IN      DS      38298 13 2 9F2D4993F47B0F2751DE0007D70A2754EE532FE373761154D9EA7A8C
B9D8EA18
br.      86400  IN      RRSIG   DS 8 1 86400 20251016170000 20251003160000 61809 . B9xwmN8hezLTALYk
l5CMV1Sum0ppa8+U7+/dp271W0mcTZZs3lQwwanL Djyf8Zw7W7jF80l10qpuqUz0A/tCrFNklsstCpfpsrRkYKH60GkDDeq rd+zGJ75V9naCot+5
q0V8s84ZJRBnrcFwzc9AXSKNsHUPf4DTERFTSLZ kzaV0+BejXEzLT2cDUu56hbiwBe00DY/nAw0VqbVQJNx9Y/HDHzy8h8 hvnIT5jg/KqNzjvb24
o/aiPwhG13kzLgXpuqheUqEwPrLnMeWL4HEPdi nm39ey0iQLk0z6FE5mS36ztxh9PZC3GShtJ8WLGevRpeilpYQoqocl2 TL7xIg==
;; Received 738 bytes from 198.97.190.53#53(h.root-servers.net) in 55 ms

;; UDP setup with 2001:12f8:a::10#53(2001:12f8:a::10) for www.unb.br failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2001:12f8:a::10#53(2001:12f8:a::10) for www.unb.br failed: network unreachable.
;; no servers could be reached
;; UDP setup with 2001:12f8:a::10#53(2001:12f8:a::10) for www.unb.br failed: network unreachable.
unb.br. 3600  IN      NS      dns1.unb.br.
unb.br. 3600  IN      NS      dns2.unb.br.
unb.br. 3600  IN      NS      ns.gigacandanga.net.br.
unb.br. 3600  IN      NS      ns2.gigacandanga.net.br.
unb.br. 3600  IN      NS      dns3.unb.br.
unb.br. 3600  IN      DS      14927 13 2 85F2FA6AEF4ED4A1F8C3B0E10D5BF1FE20D1B939A9E35F7A25280807
0A4D792B
unb.br. 3600  IN      DS      14927 13 1 0F4DF778FFC44A2D67CFF1F3575436B5F499841F
unb.br. 3600  IN      RRSIG   DS 13 2 3600 20251015235510 20251001225510 56618 br. hx0B07IRbc3rV0
90ddLCRshNMePdKHPj2G2aMWSE7SdKj6pD61GkMpKE 1TS7J7Me8Whyd1CIkRfTdyZdheGBQg==
;; Received 406 bytes from 200.192.233.10#53(c.dns.br) in 11 ms

www.unb.br. 5  IN      A      164.41.102.70
www.unb.br. 5  IN      RRSIG   A 13 3 5 20251016000000 20250925000000 14927 unb.br. ZR4Ux/NIGWIYXj
8HuqHgR8fYRHAONz4bfbkZM7MTsSkun/Mqdmnx45+4 x1vpe+CjpfZFCc3cLYXpYlgINpAWYQ==
;; Received 157 bytes from 164.41.101.6#53(dns2.unb.br) in 11 ms

```

## Parte 4 – Ferramentas alternativas

### Ferramenta 4: host

1. Resolução simples:

```

rizzonly@DESKTOP-G55DH5D:~$ host www.unb.br
www.unb.br has address 164.41.102.70

```

2. Consultar MX:

```

rizzonly@DESKTOP-G55DH5D:~$ host -t MX gmail.com
gmail.com mail is handled by 20 alt2.gmail-smtp-in.l.google.com.
gmail.com mail is handled by 5 gmail-smtp-in.l.google.com.
gmail.com mail is handled by 30 alt3.gmail-smtp-in.l.google.com.
gmail.com mail is handled by 40 alt4.gmail-smtp-in.l.google.com.
gmail.com mail is handled by 10 alt1.gmail-smtp-in.l.google.com.

```

3. Reverse lookup (PTR):

```

rizzonly@DESKTOP-G55DH5D:~$ host 8.8.8.8
8.8.8.8.in-addr.arpa domain name pointer dns.google.

```

## Ferramenta 5: drill

### 1. Consulta simples:

```
rizzonly@DESKTOP-G55DH5D:~$ drill www.google.com
;; ->>HEADER<<- opcode: QUERY, rcode: NOERROR, id: 3830
;; flags: qr rd ra ; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0
;; QUESTION SECTION:
;; www.google.com.      IN      A

;; ANSWER SECTION:
www.google.com. 300      IN      A      172.217.172.4

;; AUTHORITY SECTION:

;; ADDITIONAL SECTION:

;; Query time: 13 msec
;; SERVER: 10.255.255.254
;; WHEN: Fri Oct 3 16:34:59 2025
;; MSG SIZE rcvd: 48
```

### 2. Consultar NS:

```
rizzonly@DESKTOP-G55DH5D:~$ drill -t NS unb.br
;; ->>HEADER<<- opcode: QUERY, rcode: NOERROR, id: 44597
;; flags: qr rd ra ; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 0
;; QUESTION SECTION:
;; unb.br.             IN      NS

;; ANSWER SECTION:
unb.br. 2360      IN      NS      dns3.unb.br.
unb.br. 2360      IN      NS      dns1.unb.br.
unb.br. 2360      IN      NS      ns2.gigacandanga.net.br.
unb.br. 2360      IN      NS      ns.gigacandanga.net.br.
unb.br. 2360      IN      NS      dns2.unb.br.

;; AUTHORITY SECTION:

;; ADDITIONAL SECTION:

;; Query time: 34 msec
;; SERVER: 10.255.255.254
;; WHEN: Fri Oct 3 16:35:23 2025
;; MSG SIZE rcvd: 133
```

## Alternativa Windows (PowerShell):

```
PS C:\Users\User> Resolve-DnsName www.unb.br

Name                                     Type  TTL  Section  IPAddress
----
www.unb.br                             A      5    Answer   164.41.102.70

PS C:\Users\User> Resolve-DnsName gmail.com -Type MX

Name                                     Type  TTL  Section  NameExchange  Preference
----
gmail.com                             MX    3600  Answer   gmail-smtp-in.l.google.com      5
gmail.com                             MX    3600  Answer   alt1.gmail-smtp-in.l.google.com 10
gmail.com                             MX    3600  Answer   alt3.gmail-smtp-in.l.google.com 30
gmail.com                             MX    3600  Answer   alt2.gmail-smtp-in.l.google.com 20
gmail.com                             MX    3600  Answer   alt4.gmail-smtp-in.l.google.com 40

PS C:\Users\User> Resolve-DnsName 8.8.8.8 -Type PTR

Name                                     Type  TTL  Section  NameHost
----
8.8.8.8.in-addr.arpa                  PTR    84077 Answer   dns.google
```

## Parte 5 – DNSSEC E TCP

### 1. Verificar DNSSEC:

```
rizzonly@DESKTOP-G55DH5D:~$ dig dnssec-failed.org +dnssec
; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> dnssec-failed.org +dnssec
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 13364
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: do; udp: 1220
;; QUESTION SECTION:
;dnssec-failed.org.          IN      A

;; ANSWER SECTION:
dnssec-failed.org.          300     IN      A          96.99.227.255
dnssec-failed.org.          300     IN      RRSIG      A 5 2 300 20251019145131 20251002144631 44973 dnssec-failed.org. JA
ImJjTbALJ+G9g8tLGK6v1u/MyzItgoCzRCEgptZr6/jVByML+7AIjK lqhle0ArYYMtpm3Z53pYFVLSgokYaRB69VUwJ4M4l/p+j1DsnJrcI9nn /Ho
XSLHz+TXJx2NgcEYbV+6qMaLW97cj7aCOTqM8KoqNxsY3wHzU2v eFY=

;; Query time: 172 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Fri Oct 03 16:37:13 -03 2025
;; MSG SIZE rcvd: 239
```

### 2. Forçar TCP em vez de UDP:

```
rizzonly@DESKTOP-G55DH5D:~$ dig www.unb.br +tcp
; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> www.unb.br +tcp
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 56010
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:;; udp: 1400
;; QUESTION SECTION:
;www.unb.br.                IN      A

;; ANSWER SECTION:
www.unb.br.                 5       IN      A          164.41.102.70

;; Query time: 27 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (TCP)
;; WHEN: Fri Oct 03 16:37:53 -03 2025
;; MSG SIZE rcvd: 55
```

## Questões de Reflexão:

### 1. Como o TTL afeta o tempo de propagação de mudanças?

O TTL afeta o tempo de propagação de mudanças de forma direta. De maneira simples, quanto mais baixo o TTL = propagação mais rápida, mais alto o TTL = propagação mais lenta. A “propagação” de DNS é o tempo que os servidores DNS intermediários levam para limpar suas informações (cache) e buscar novas. O TTL está diretamente ligado a agilidade das mudanças no DNS, sendo a ferramenta que dá controle para tal.

### 2. Qual a diferença entre consulta recursiva e iterativa?

Na consulta recursiva, o meu computador faz uma consulta recursiva ao servidor DNS, que por sua vez recebe o pedido sendo obrigado a retornar a resposta final. Ele que irá se “virar” para fazer as consultas iterativas aos outros servidores em meu nome. Já na consulta iterativa, os servidores DNS fazem a consulta entre si. O servidor que recebe o pedido não é obrigado a ter a resposta final. Ele apenas dá a melhor “pista” que tem, que é um encaminhamento para outro servidor DNS mais específico.

3. Por que resolvedores públicos podem ser mais rápidos?

Por atenderem bilhões de pessoas, a chance do site que nos queremos acessar já estar no cache deles é alta, tornando a resposta muito rápida. Além disso, eles possuem servidores espalhados pelo mundo todo. A consulta é automaticamente enviada para o servidor mais próximo, o que diminui a latência.

4. Para que serve o reverse lookup (PTR)?

A consulta reversa (PTR) serve para descobrir o nome de um domínio a partir do endereço de IP do usuário. Ele é muito utilizado pelo efeito anti-spam, ou seja, servidores de email verificam o nome do IP de origem para ajudar a validar se o remetente é legítimo. E também, para logs, transformando IPs em nomes legíveis em relatórios e logs de rede, facilitando a identificação da origem de um acesso.